

■ Electro, Physical & Theoretical Chemistry

Improvement of Cycleability and Rate-Capability of $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ Cathode Materials Coated with Lithium Boron Oxide by an Antisolvent Precipitation MethodSatoshi Hashigami,^{*,[a], [b]} Kei Yoshimi,^[b] Yukihiro Kato,^[b] Hiroyuki Yoshida,^[a] Toru Inagaki,^[a] Masakazu Haruta,^[b] Michihiro Hashinokuchi,^[b] Takayuki Doi,^[b] and Minoru Inaba^[b]

$\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ (NCM) particles coated with lithium boron oxide were prepared by an antisolvent precipitation method. In the antisolvent precipitation method, ethanol was used to strip off the hydrated water coordinated with lithium ions and borate ions. By the antisolvent precipitation method, the NCM particles were coated with a uniform layer of LiBO_2 . The capacity fading on cycling was successfully suppressed for the lithium boron oxide-coated samples. In addition, the rate capability was also improved by the coating. The lithium boron

oxide coating effectively suppressed the increase of the electrode impedance associated with the electrolyte decomposition. Crack formation in the secondary particles after charge-discharge cycling was greatly inhibited for lithium boron oxide-coated NCM. It was demonstrated that the lithium boron oxide coating layer suppressed the side reactions not only on the particle surface, but also in the intergranular cracks by inhibiting the penetration of the electrolyte solution.

Introduction

Layered transition metal oxides $\text{LiNi}_{1-x-y}\text{Co}_x\text{Mn}_y\text{O}_2$ have attracted much attention as cathode materials of lithium ion batteries (LIBs). $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ is one of the most promising compositions because it provides a higher capacity (ca. 200 mAh g^{-1} when charged to 4.6 V vs. $\text{Li}^+/\text{Li}^{[1-3]}$) than LiCoO_2 and $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$ that are currently used in commercially available LIBs.^[3-5] However, it is known that the high potential charging of $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ results in poor cycle performance.^[6] Studies in the literature have so far revealed several important factors for the capacity fading of the layered materials, which include the decomposition of electrolytes through side reactions of LiPF_6 with water molecules^[5-7] and structural transformations to a mixed spinel/rock-salt structure on the individual particle surface.^[7,8] The dissolution of transition metal ions from $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ is also considered to be an important degradation factor.^[9] In addition to the chemical instability, crack formation within the secondary particles is well-known as a degradation factor under high potential operation.^[10,11] The cracks disconnect the electronic pathway among the grains in a secondary particle. The irreversible reaction of electrolyte decomposition occurs not only on the

surface, but also inside the cracks, which accelerates crack formation and capacity fading.

As an approach to suppress direct contact between the cathode materials and the electrolyte solution, surface coating with various oxides such as Al_2O_3 ,^[12] ZrO_2 ,^[13] TiO_2 ,^[14] CeO_2 ,^[15] and Nb_2O_5 ^[16] has been investigated. In addition, there have been several reports of doubly coating or doping the active material with two components.^[17,18] Recently, we reported that the homogeneous coating layer of lithium silicate formed by an antisolvent precipitation method not only covered the surface of $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ particles, but also penetrated the grain boundaries, which effectively suppressed intergranular crack formation on cycling and improved cycleability.^[19] However, the lithium silicate coating retarded the rate-capability. Furthermore, the amount of lithium silicate coating was limited up to 0.5 wt%, because further addition remarkably decreased the capacity owing to a poor Li^+ ion conductivity of lithium silicate. Lithium boron oxide is a Li^+ conductor though the conductivity is not so high (e.g. $10^{-7} \text{ S cm}^{-1}$ for LiBO_2 ^[20]). Wang et al. used lithium boron oxide having a high ion conductivity as a coating material for $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$, which successfully reduced the over-potential at the cathode particle/electrolyte interface.^[21] However our preliminary study revealed that $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ cathode coated with lithium boron oxide by a conventional impregnation method did not remarkably improve the cycle performance.

In this study, we coated $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ (NCM) with lithium boron oxide (LBO) by an antisolvent precipitation method, which we developed for a uniform coating layer of lithium silicate (LSO) in a previous study,^[19] and investigated the charge and discharge characteristics of NCM cathode. The morphology and composition of the LBO coating and its effects on the chemical and physical stability of NCM are discussed.

[a] Dr. S. Hashigami, Dr. H. Yoshida, Dr. T. Inagaki
Fundamental Technology Laboratory, R&D Center, The Kansai Electric Power Co., Inc., 11–20 Nakoji 3-Chome, Amagasaki, Hyogo 661–0974, Japan
E-mail: hashigami.satoshi@c5.kepc.co.jp

[b] Dr. S. Hashigami, K. Yoshimi, Y. Kato, Dr. M. Haruta, Dr. M. Hashinokuchi, Prof. T. Doi, Prof. M. Inaba
Department of Molecular Chemistry and Biochemistry, Doshisha University, 1–3 Tataramiyakodani, Kyotanabe, Kyoto 610–0321, Japan

Supporting information for this article is available on the WWW under <https://doi.org/10.1002/slct.201900874>

Results and Discussion

XRD patterns of uncoated and LBO-coated NCM samples are shown in Figure 1. The XRD pattern of each sample was typical

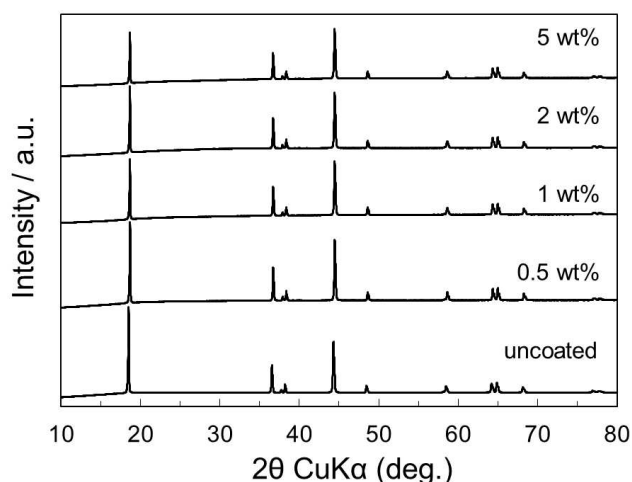


Figure 1. XRD patterns of uncoated and LBO-coated NCM samples prepared by the antisolvent precipitation method and heat-treated at 500 °C for 8 h.

of the layered structure of NCM. Crystalline LBO was not detected by XRD, which indicated that LBO may exist as amorphous state. The peak intensity ratio R of the (003) to the (104) peak is known to be a measure of the cation mixing.^[6] The R value decreased with increasing the amount of LBO modification. Although the reason is not clear, the increase of the water amount used to dissolve LiBO_2 in the first step in the antisolvent precipitation method may affect the dissolution of lithium from the active materials, resulting in the observed increase in cation mixing. In addition, Han et al. reported that this phenomenon is attributed to partial occupation of small-sized boron ions in lithium ion positions within the layered structure.^[22] Therefore, in the antisolvent precipitation method, the addition of an excess amount of LiBO_2 is harmful with respect to cation mixing.

Figure 2 shows SEM images of the uncoated and the LBO-coated $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ particle surfaces. No appreciable difference in surface morphology was observed among the uncoated NCM, 0.5 wt% LBO-NCM and 2 wt% LBO-NCM. On the other hand, the surface of 5 wt% LBO-NCM showed that the grain boundaries of the primary particles became unclear and the modification with LBO seems to be excessive. Figure 3 shows AES elementary maps of B + Ni, B and Ni of 2 wt% LBO-NCM. Boron was uniformly distributed on the 2 wt% LBO-NCM particle surface, which clearly confirmed thin and uniform coating of LBO on the particle surface.

In order to estimate the composition and crystal structure of LBO coating layer deposited by the antisolvent precipitation method, only the coating material (without $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$) was synthesized via the antisolvent precipitation method and calcined at 500 °C for 8 h. The XRD pattern of the resulting

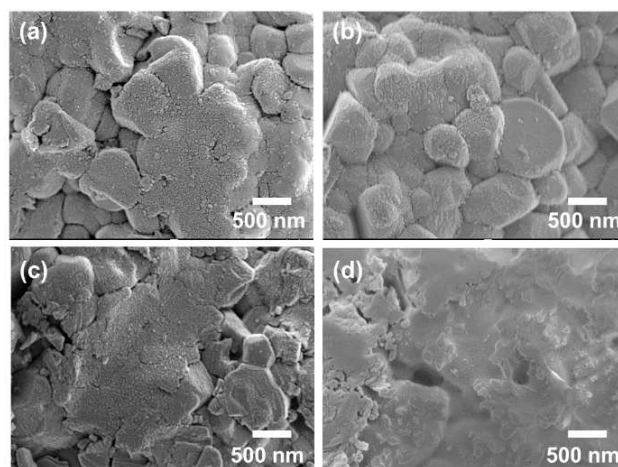


Figure 2. SEM images of (a) uncoated NCM, (b) 0.5, (c) 2, and (d) 5 wt% of LBO-NCM prepared by the precipitation method.

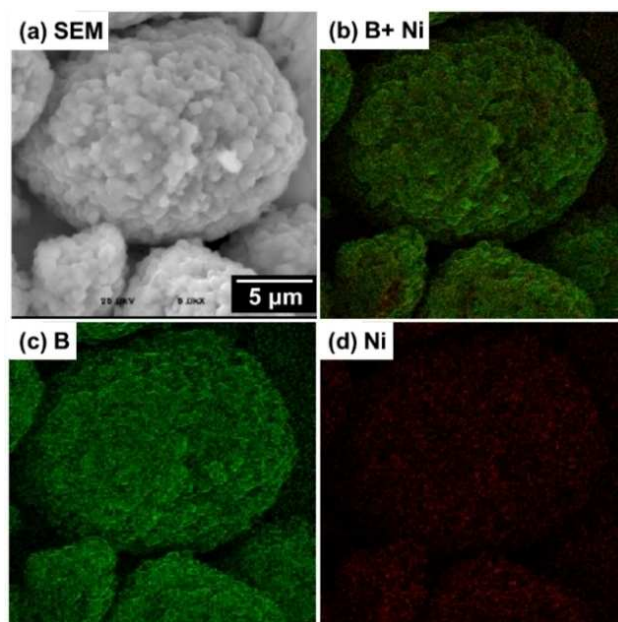


Figure 3. SEM images (a) and AES elementary maps of (b) B + Ni, (c) B, and (d) Ni of 2 wt% LBO-NCM.

powder is shown in Figure 4. The diffraction pattern of the coating material prepared by the antisolvent precipitation method was assigned to that of LiBO_2 . In the antisolvent precipitation method, ethanol was used to strip off the hydrated water coordinated with lithium ions and borate ions. This treatment promotes the reaction of free lithium ions and borate ions to give LiBO_2 precipitation. Since the coating material is precipitated from a large amount of solution, a uniform coating layer of LiBO_2 single phase can be formed on the surface of NCM particles.

LBO coating has also been used to modify the surfaces of $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$,^[21] $\text{LiNi}_{1/3}\text{Co}_{1/3}\text{Mn}_{1/3}\text{O}_2$,^[23] and $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$.^[24]

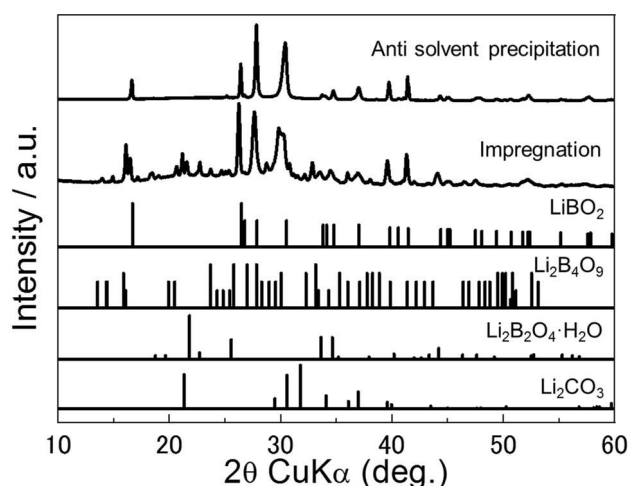


Figure 4. XRD patterns of the coating material (without NCM particles) prepared by the antisolvent precipitation method and the impregnation method and heat-treated at 500 °C for 8 h.

In these reports, LBO was coated on the cathode particles using conventional impregnation methods; however, the crystal structure of the boron coating material has not been identified. The XRD diffraction pattern of the coating material obtained by drying aqueous LiBO_2 solution at 80 °C, followed by calcination at 500 °C for 8 h, which reproduced the conditions of the impregnation method, is also shown in Figure 4. The diffraction pattern is assigned to a mixture of $\text{LiB}_2\text{O}_4\cdot\text{H}_2\text{O}$, $\text{Li}_2\text{B}_4\text{O}_9$, and LiBO_2 . Because the composition of the precipitates is varied during the drying process in the impregnation method because of different solubilities of LBOs ($\text{LiB}_2\text{O}_4\cdot\text{H}_2\text{O}$, $\text{Li}_2\text{B}_4\text{O}_9$, and LiBO_2) in water, it may be difficult to obtain a single phase. Furthermore, it has been reported that the lithium-ion conductivity of $x\text{Li}_2\text{O}\cdot\text{B}_2\text{O}_3$ glass increases with the amount of Li (x) up to ca. $x=3.5$ at room temperature.^[25] Therefore, the lithium ion conductivity of the coating layer deposited by the antisolvent precipitation method, which mainly consisted of LiBO_2 , is expected to be higher than the mixture of LiBO_2 , $\text{LiB}_2\text{O}_4\cdot\text{H}_2\text{O}$ and $\text{Li}_2\text{B}_4\text{O}_9$ because LiBO_2 has a higher Li/B ratio than the other two borates. In addition, the hydrate $\text{Li}_2\text{B}_2\text{O}_4\cdot\text{H}_2\text{O}$ introduces moisture into the cell, and may lead to the formation of HF in the cell.

Figs. 5(a), (b) and (c) show the charge and discharge profiles at 0.1 C between 2.5 and 4.6 V at 30 °C for the uncoated NCM, and 0.5 and 2 wt% LBO-NCM, respectively. All samples provided discharge capacities of around 200 mAh g^{-1} at the initial cycles, which are typical values of the layered NCM cathodes when charged to 4.5–4.6 V.^[1–3] These data showed that the LBO coating layer prepared by the antisolvent precipitation method does not inhibit lithium ion transport. Figure 5(d) shows the variations of the specific capacity with cycle number for the uncoated NCM, 0.5 and 2.0 wt% LBO-NCM and 0.5 wt% LSO-NCM.^[19] The capacity fading on cycling was successfully suppressed for the LBO- and LSO-coated samples. The capacity retentions at the 100th cycle of 0.5 wt% LBO-NCM (74.9%), 2 wt% LBO-NCM (78.5%) and 0.5 wt% LSO-NCM (76.6%) were

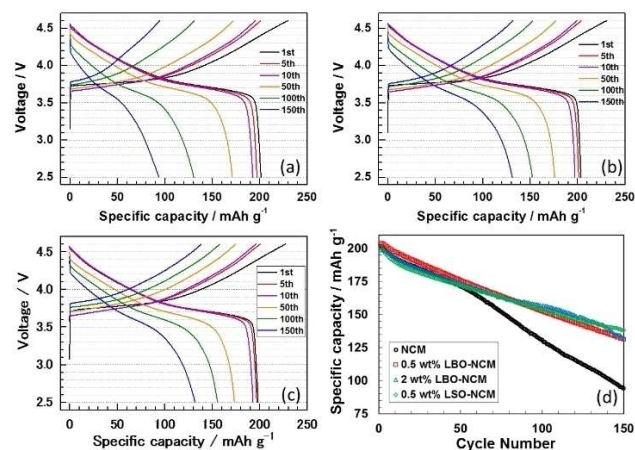


Figure 5. Charge-discharge curves of (a) uncoated NCM, (b) 0.5 wt% LBO-NCM, and (c) 2 wt% LBO-NCM, and (d) capacity retentions measured at 0.1 C between 2.5 and 4.6 V in 1 M $\text{LiPF}_6/\text{EC} + \text{DMC}$ (1:2) at 30 °C.

almost the same. Even 2 wt% LBO-NCM with a higher coating amount gave a high capacity and good cycle performance, whereas this was not the case for LSO coating in a previous study.^[19] This is owing to a lithium ion conductivity of LBO higher than LSO.

Figure 6 shows the rate capability of the uncoated NCM, 0.5, 2 and 5 wt% LBO-NCM and 0.5 wt% LSO-NCM. The 0.5 and

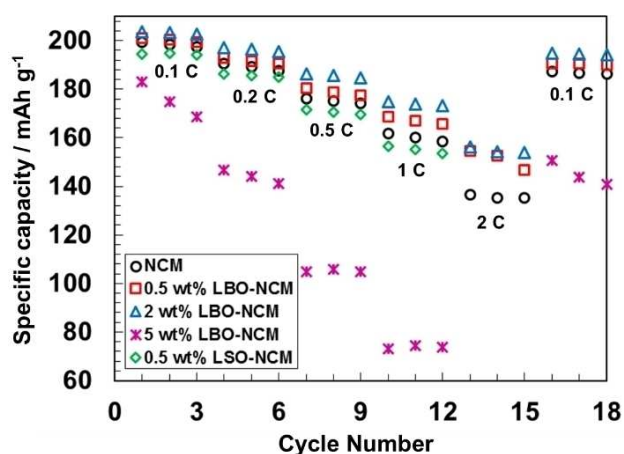


Figure 6. Rate-capability of uncoated NCM, 0.5, 2 and 5 wt% LBO-NCM, and 0.5 wt% LSO-NCM between 2.5 and 4.6 V in 1 M $\text{LiPF}_6/\text{EC} + \text{DMC}$ (1:2) at 30 °C.

2 wt% LBO-NCM exhibited better rate capability than the uncoated NCM, and the discharge capacity of 2 wt% LBO-NCM at 2 C was by 20 mAh g^{-1} higher than the uncoated NCM. In the case of 5 wt% LBO-NCM, the discharge capacity decreased rapidly at high rates. This deterioration was attributable to an increase in the interfacial resistance owing to an excess amount of the coating material as shown in Figure 2(d). Thus, we concluded that the optimum coating amount of LBO was 2 wt

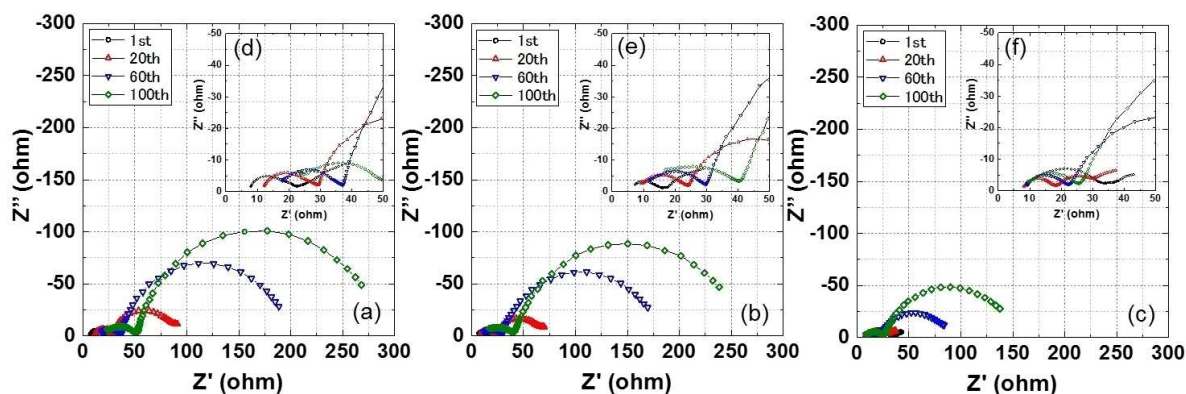


Figure 7. Nyquist plots of (a) uncoated NCM, (b) 0.5 wt% LBO-NCM and (c) 2 wt% LBO-NCM measured at 3.8 V after the 1st, 20th, 60th, 100th cycle between 2.5 and 4.6 V at 30 °C. Plots (d), (e) and (f) are magnified ones of (a), (b) and (c), respectively.

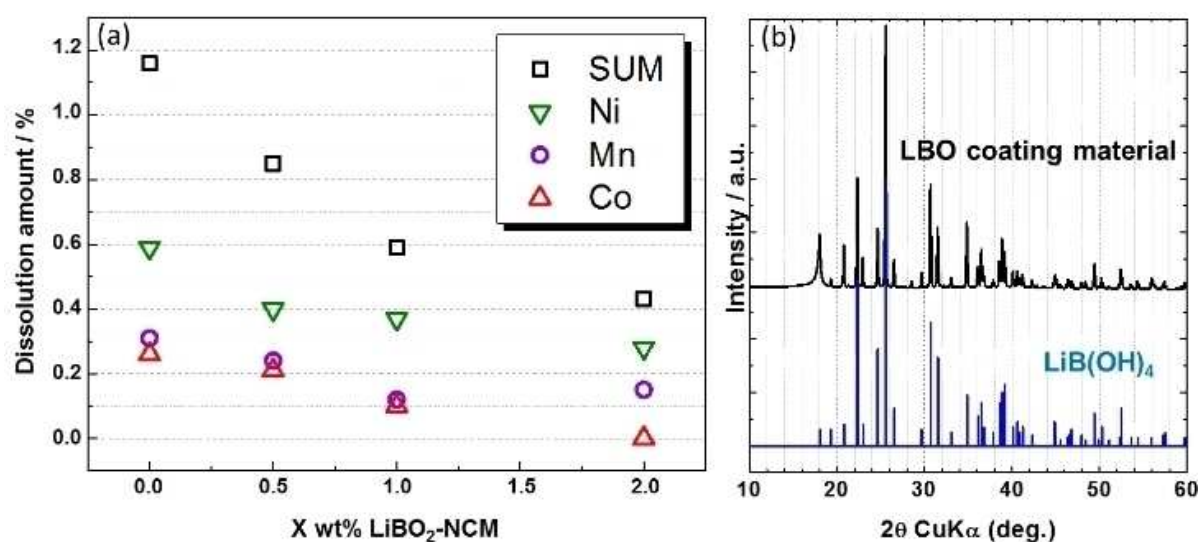


Figure 8. (a) Amounts of dissolved transition metal ions from the uncoated NCM, 0.5, 1 and 2 wt% LBO-NCM after 10 cycles at 0.5 C between 2.5 and 4.6 V at 30 °C and stored at the fully charged state at 45 °C for 120 h. (b) XRD patterns of LiB(OH)₄ and the LBO coating material prepared by the antisolvent precipitation method after soaked in a mixture of the electrolyte (20 ml) for 20 days in the air.

%. On the other hand, 0.5 wt% LSO-NCM showed poorer rate-capability than the uncoated NCM most probably because of a low lithium ion conductivity of LSO.^[26] These results suggest that suitable choice of coating material with an optimized amount not only suppresses side reactions on the cathode particle surface, but also accelerates the interfacial reactions at the electrode/electrolyte interface. A high lithium ion conductivity seems to be an important factor to enhance the interfacial reactions.

Electrochemical impedance spectroscopy was carried out to investigate the interfacial electrochemistry. Figure 7 shows Nyquist plots of the uncoated NCM, and 0.5 and 2 wt% LBO-NCM at 3.8 V after the 1st, 20th, 60th, 100th cycle. In each spectrum, two semi-circles were observed. Generally, the intersection between the real axis and the higher frequency arc is attributed to the ohmic resistance of the electrolyte solution. The higher-frequency arc is assigned to the contact resistance

among the active material and conducting carbon particles. The lower-frequency arc is assigned to the charge transfer resistance (lithium ion transfer resistance) at the electrode/electrolyte interface. The increase of both the contact resistance and the charge-transfer resistance during cycling was suppressed by LBO coating; especially the suppression of the latter was remarkable for 2.0 wt% LBO-NCM. These results indicate that the LBO coating inhibited the oxidative decomposition of the electrolyte on the particle surface and the resulting inactivation of the particle surface, which is the reason for the improved rate-capability shown in Figure 6.

As mentioned earlier, dissolution of transition metal ions from the cathode is one of the reasons for capacity fading. Hence the effect of LBO-coating on transition metal ion dissolution was investigated. Figure 8(a) shows the amounts of dissolved transition metal ions at 45 °C for 120 h plotted against the amount of coated LBO. All the three transition

metals were dissolved more or less at 45 °C, and the highest amount was Ni followed by Mn and Co. The presence of LBO coating at the surface was significantly effective for suppression of the dissolution, and the total amount of dissolved ions was reduced to 37% for 2 wt% LBO-NCM.

In order to investigate the reactivity of LBO with impurities in the electrolyte, 1 g of the coating material (without $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$) prepared by the antisolvent precipitation method was soaked in a mixture of the electrolyte (20 mL), and stirred at 45 °C for 20 days being exposed to the air as an accelerated test. The solution was filtered and the XRD pattern of the residue was measured, which is shown in Figure 8(b). $\text{LiB}(\text{OH})_4$ was a main compound in the residue, which was formed by a reaction of H_2O with LiBO_2 . It was suggested that the presence of the coating material effectively traps trace H_2O in the electrolyte solution, which may inhibit HF formation from LiPF_6 . It is considered that the LBO coating not only inhibits direct contact with the electrolyte, but also suppresses side reactions caused by trace water (and the resulting HF), which resulted in the suppression of the dissolution of transition metal ions.

Figure 9 shows cross-sectional SEM images of the uncoated NCM and 2 wt% LBO-NCM obtained before and after long-term cycling tests. The tests were performed at 0.5 C between 2.5 and 4.6 V, while the reversible capacity was checked every 20 cycles at 0.1 C. Before cycling, the particle consisted of small and tightly aggregated primary particles of NCM. However, many cracks were formed along the grain boundaries of the secondary particles after 106 cycles for the uncoated sample, and crack formation was further extended after 211 cycles. On the other hand, crack formation was much less significant even after 211 cycles for the 2 wt% LBO-NCM. The electrolyte solution can penetrate into the intergranular cracks inside the particles. This may cause irreversible electrolyte decomposition within the particles. As a result, the decomposition products accumulated in the intergranular cracks and accelerated the capacity fading as was discussed in a previous study.^[19] On the other hand, the generation of cracks was clearly suppressed for the LBO-coated sample. It is considered that the LBO coating suppressed the progress of the cracks probably by inhibiting the penetration of the electrolyte solution into the intergranular cracks.

Conclusions

The surface of $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ was coated with LBO by an antisolvent precipitation method to improve its electrochemical properties such as cycleability and rate-capability. By the antisolvent precipitation method, the $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ particles were coated with a uniform layer of LiBO_2 . The LBO layer not only suppressed the capacity fading on cycling, but also improved the rate-capability. The optimum value of the amount of the coating material was 2 wt%. We found that the dissolution of transition metal ions and the increase of impedance are remarkably reduced by the LBO coating, which clearly indicated that uniform LBO layer effectively covered the NCM particle surface and suppressed metal ion dissolution and

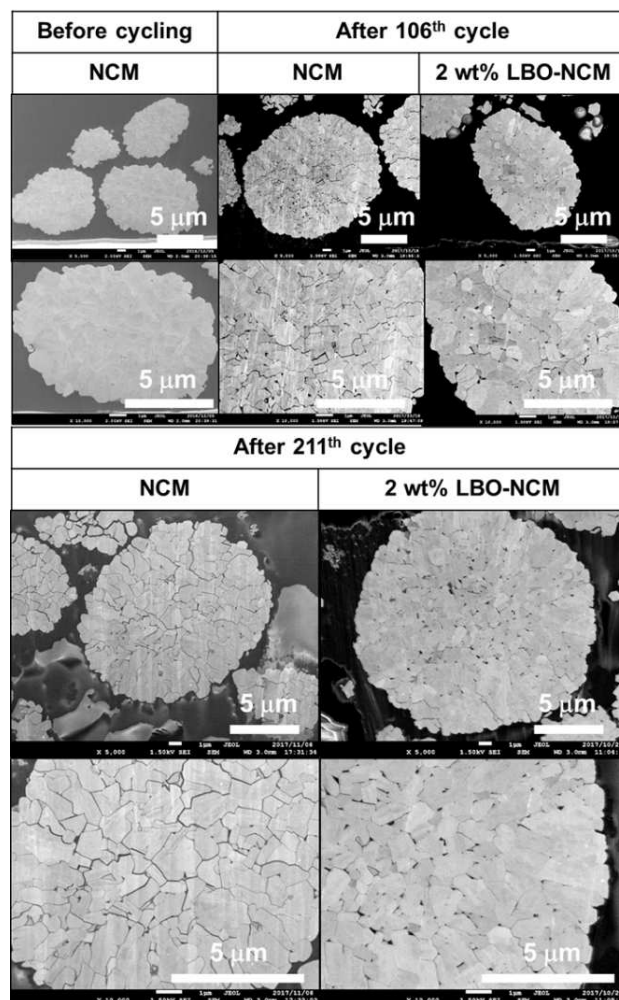


Figure 9. Cross sectional SEM images of uncoated NCM and 2 wt% LBO-NCM particles before cycling, and after 106 and 211 cycles at 0.5 C between 2.5 and 4.6 V in 1 M $\text{LiPF}_6/\text{EC} + \text{DMC}$ (1:2) at 30 °C.

solvent decomposition at high potentials. The Li^+ conductivity of LBO higher than LSO also contributed to the improved rate-capability. In addition, the LBO-coating suppressed crack formation along the grain boundaries of the secondary particles by inhibiting the penetration of the electrolyte solution into the intergranular cracks. It is therefore concluded that LBO coating by the antisolvent precipitation method is an easy and effective way for improving the cyclability and rate capability of $\text{LiNi}_{0.5}\text{Co}_{0.2}\text{Mn}_{0.3}\text{O}_2$ cathode.

Supporting Information Summary

Supporting information is available from the Wiley Online Library. Experimental methods, Figure S1.

Conflict of Interest

The authors declare no conflict of interest.

Keywords: Batteries • Crack Formation • Electrodes • Nickel-rich Layered Cathodes • Surface Chemistry

- [1] P. Yan, J. Zheng, M. Gu, J. Xiao, J.-G. Zhang, C.-M. Wang, *Nat. Commun.* **2017**, *8*, 14101.
- [2] K. Ishidzu, Y. Oka, T. Nakamura, *Solid State Ionics* **2016**, *288*, 176–179.
- [3] S. K. Jung, H. Gwon, J. Hong, K.-Y. Park, D.-H. Seo, H. Kim, J. Hyun, W. Yang, K. Kang, *Adv. Energy Mater.* **2014**, *4*, 1300787.
- [4] M. Noh, J. Cho, *J. Electrochem. Soc.* **2013**, *160*(1), A105–A111.
- [5] M. Borner, F. Horsthemke, F. Kollmer, S. Haseloff, A. Friesen, P. Niehoff, S. Nowak, M. Winter, F. M. Schappacher, *J. Power Sources* **2016**, *335*, 45–55.
- [6] H.-J. Noh, S. Youn, C. S. Yoon, Y.-K. Sun, *J. Power Sources* **2013**, *233*, 121–130.
- [7] F. Lin, L. M. Markus, D. Nordlund, T.-C. Weng, M. D. Asta, H. L. Xin, M. M. Doeff, *N. Commun.* **2014**, *5*, 3529.
- [8] C.-H. Shen, Q. Wang, H.-J. Chen, C.-G. Shi, H.-Y. Zhang, L. Huang, J.-T. Li, S.-G. Sun, *ACS Appl. Mater. Interfaces* **2016**, *8*, 35323–35335.
- [9] J. A. Gilbert, I. A. Shkrob, D. P. Abraham, *J. Electrochem. Soc.* **2017**, *164*(2), A389–A399.
- [10] J. Li, L. E. Downie, L. Ma, W. Qiu, J. R. Dahn, *J. Electrochem. Soc.* **2015**, *162*(7), A1401–A1408.
- [11] H. Liu, M. Wolf, K. Karki, Y. -S. Yu, E. A. Stach, J. Cabana, K. W. Chapman, P. J. Chupas, *Nano Lett.* **2017**, *17*, 3452–3457.
- [12] Y. Shi, M. Zhang, D. Qian, Y. S. Meng, *Electrochim. Acta* **2016**, *203*, 154–161.
- [13] J.-Z. Kong, S.-S. Wang, G.-A. Tai, L. Zhu, L.-G. Wang, H.-F. Zhai, D. Wu, A.-D. Li, H. Li, *J. Alloys Comp.* **2016**, *657*, 593–600.
- [14] W. Liu, M. Wang, X. L. Gao, W. Zhang, J. Chen, H. Zhou, X. Zhang, *J. Alloys Comp.* **2012**, *543*, 181–188.
- [15] K. Liu, G.-L. Yang, Y. Dong, T. Shi, and L. Chen, *J. Power Sources* **2015**, *281*, 370–377.
- [16] P. Karayaylali, R. Tatara, Y. Zhang, K. L. Chan, Y. Yu, L. Giordano, F. Maglia, R. Jung, I. Lund, Y. S. Horn, *J. Electrochem. Soc.* **2019**, *166*(6), A1022–A1030.
- [17] T. Gu, J. Wang, J.-H. Tian, X. Zheng, K. Lu, Y. Xin, H. Wang, R. Yang, *ChemElectroChem* **2019**, *6*, 2224–2230.
- [18] B. Seteni, N. Rapulenyane, J. C. Ngila, S. Mpelane, H. Luo, *J. Power Sources* **2017**, *353*, 210–220.
- [19] S. Hashigami, Y. Kato, K. Yoshimi, H. Yoshida, T. Inagaki, M. Hashinokuchi, T. Doi, M. Inaba, *Electrochim. Acta* **2018**, *291*, 304–310.
- [20] Y. Ito, K. Miyauchi, T. Oi, *J. Non-Crystalline Solids* **1983**, *57*, 389–400.
- [21] D. Wang, X. Li, Z. Wang, H. Guo, X. Chen, X. Zheng, Y. Xu, J. Ru, *Electrochim. Acta* **2015**, *174*, 1225–1233.
- [22] S.-H. Han, J. H. Song, T. Yim, Y.-J. Kim, J.-S. Yu, S. Yoon, *J. Electrochem. Soc.* **2016**, *163*(5), A748–750.
- [23] S. Tan, L. Wang, L. Bian, J. Xu, W. Ren, P. Hu, A. Chang, *J. Power Sources* **2015**, *277*, 139–146.
- [24] J. S. Chae, S.-B. Yoon, W.-S. Yoon, Y.-M. Kang, S.-M. Park, J.-W. Lee, K. C. Roh, *J. Alloys Comp.* **2014**, *601*, 217–222.
- [25] E. E. Horopanitis, G. Perentzis, E. Pavlidou, L. Papadimitriou, *IONICS* **2003**, *9*, 88–94.
- [26] I. D. Raistrick, C. Ho, R. A. Huggins, *J. Electrochem. Soc.* **1976**, *123*(10), 1469–1476.

Submitted: March 7, 2019

Accepted: July 24, 2019